

Sphere Packing: How optimally stacking cannonballs strains human problem-solving

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Abstract

Sphere packing is the study of how to place spheres as tightly as possible in space. Although we can easily state the problem, intuitive facts can be excruciatingly difficult to prove. In 1611, Kepler correctly conjectured that a pyramid configuration has maximal density; however, it wasn't proven until 1998 due to Hales. Still, the proof was almost 300 pages and required extensive computer calculations. Even the simpler 2-dimensional problem wasn't rigorously proven until 1940.

In this paper, we outline of the state of the sphere packing problem. We begin by reviewing some history and applications of sphere packing. Then we rigorously define the problem. Afterwards, we provide a proof of the maximum density in 2-dimensions and an overview of Hales' proof of the maximum density in 3-dimensions. Next, we generalize the problem to higher dimensions by generalizing the notions of spheres and volumes. We discuss some lower bounds on the density in several dimensions. Lastly, we define lattices, which have been used to construct many of the best-known packings. There are extremely symmetrical lattice packings in 8 and 24 dimensions, and researchers have proven that they have maximal densities. We provide constructions of the lattices and the intuition behind the proof that their densities are maximal. Sphere packing is still an active area of research, and the maximum density in all other dimensions is unknown.

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1 Introduction

1.1 History

Sphere packing is the problem of placing spheres as densely as possible into space. If you have ever seen fruit stacked at the grocery store, you can imagine one possible packing, which is similar to the Face-Centered Cubic (FCC) packing depicted in Figure 1.

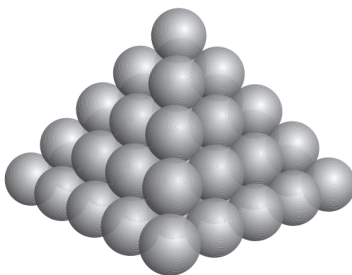


Figure 1: The FCC Packing [1].

The study of the mathematical properties of the FCC packing can be traced back to a Sanskrit work composed in 499 CE. More recently, mathematician Thomas Harriot began to study sphere configurations after his correspondence with Sir Walter Raleigh in 1606. Raleigh had given Harriot the problem of determining the formulas for the number of cannonballs in regularly stacked piles. Harriot found that the formula for stacking cannonballs is the same as the formula for the tetrahedral numbers. In fact, Harriot connected the problem to Pascal's Triangle long before Pascal studied it (Figure 2)[1].

Soon afterwards, mathematicians became interested in the modern problem of sphere packing; they wanted to find the densest or tightest possible packing of spheres. Intuitively, we might believe that the FCC packing is the densest arrangement. Johannes Kepler thought so too, and in 1611 he published the statement as a conjecture. He also stated that if we fill a container with spheres, the volume of the spheres over the volume of the container is at most $\pi/\sqrt{18}$ [1]. Although this is a simply stated conjecture, it took nearly 400 years to prove! Even the maximum density in the simpler 2-dimensional problem wasn't proven rigorously until 1940 by László Fejes Tóth [2]. In 3-dimensions, there was progress as people improved bounds on the density, found classes of packings for which the bound of $\pi/\sqrt{18}$ on the density was known, studied convex bodies—other than spheres—for which the density can be determined precisely, and tested proof strategies. But Kepler's conjecture wasn't proven until Thomas Hales verified it computationally in 1998 [3].

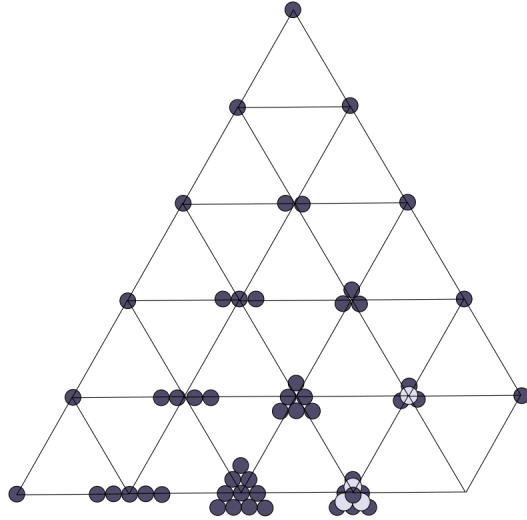


Figure 2: In this version of Pascal's Triangle, the number of spheres corresponds to the standard entries. The n -th diagonal starting on the right side represents stackings of cannonballs in n -dimensions [1].

1.2 Applications

To many, the sphere packing problem is inherently interesting, but it also has many applications. The 2-dimensional circle packing problem is useful for cell phone coverage, packing, tree farming, and cutting. Cell phone companies have cell towers that reach over a certain range and provide service within a circle around them. To maximize coverage without redundantly placing towers, companies must place them, so the covered ranges are as dense as possible without overlapping. Companies that pack bottles or other cylindrical materials must focus on the placement of circular bottoms; it's important for them to pack the circles as densely as possible. Tree farmers need to optimize where they plant each tree. From a bird's-eye-view, trees look like circles. If each tree grows to about the same radius, farmers can plant their seeds so that each tree can grow to its full size without wasting space or inhibiting the growth of other trees. Lastly, industries who need to cut circles out of larger materials must choose where to cut to minimize waste [4].

The 3-dimensional sphere packing problem is useful because it serves as a model for studying granular materials, and it is a baseline for packing more complex materials. Furthermore, high-dimensional sphere packings are often used in signal processing. Signals can typically be represented with high-dimensional vectors. For example, a radio signal is a superposition of waves, which can be represented with a vector—where each component represents the amplitude of a specific frequency in the superposition. Generally, the signal sent $\mathbf{s}$ is slightly different from the one received $\mathbf{r}$ because of noise in the channel, but they only differ by some distance ϵ . We can think

of the signal as a sphere of radius ϵ around $\mathbf{s}$ (Figure 3). If error spheres overlap, two signals could get confused (Figure 4).

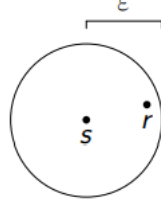


Figure 3: Our signal $\mathbf{s}$ can be received as any signal $\mathbf{r}$ within the error ball [5].

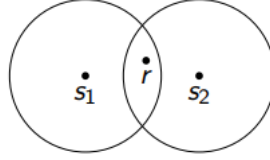


Figure 4: If the error balls of two signals $\mathbf{s}_1$ and $\mathbf{s}_2$ overlap, and we receive a signal $\mathbf{r}$ that is in both balls, we can't determine which signal was sent [5].

To avoid ambiguous signals, we can restrict signals we send. The possible signals must form a sphere packing. This is called an *error correcting code* [5].

1.3 A Rigorous Statement of Kepler's Conjecture

Before we discuss some of the major proofs in sphere packing, we need to establish the problem more rigorously. All definitions below are discussed in [1].

Definition 1.1 (Sphere Packings) *A sphere packing is an arrangement of non-overlapping spheres of radius 1 in Euclidean space. Equivalently, since each sphere's position is determined by its center, a packing of spheres is a set of points such that every pair of points is at least distance 2 apart in Euclidean space.*

Definition 1.2 (Saturated Sphere Packings) *A sphere packing—the point set of the centers of the spheres—is saturated if it is not a proper subset of any other sphere packing.*

Definition 1.3 (Density of a Sphere Packing) *Consider a large sphere container of radius R filled with unit spheres. The density of the sphere packing is given by*

$$\limsup_{R \rightarrow \infty} \frac{\text{volume of spheres}}{\text{volume of the space}}.$$

Since the density is defined as a limit, it doesn't matter what shape we consider to fill with our sphere packing. We could imagine filling boxes, ellipsoids, or any other shape with spheres.

Definition 1.4 (Face Centered Cubic or Cubic Close Packing) *A face centered cubic (FCC) packing or cubic close packing (CCP) is a packing such that if we partition Euclidean space into cubes of side-length $2\sqrt{2}r$, where r is the radius of each sphere, the centers of each sphere fall exactly on the vertices and centers of each face of the cubes (Figure 5).*

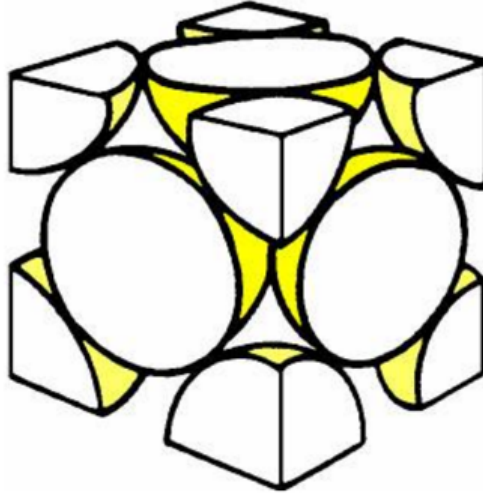


Figure 5: This image from [4] shows how to construct the FCC packing. When we partition Euclidean space into cubes, the centers of the spheres fall on the vertices and the centers of the faces. Notice that since the diagonal of a face is $4r$, the length of a side must be $2\sqrt{2}r$.

We can easily compute the density of the FCC packing. Notice that there are eight eighths of spheres on the corners and six halves of spheres on the faces in the cube, giving us $1 + 3 = 4$ total spheres. Therefore, the area of the spheres is

$$4\left(\frac{4}{3}\pi r^3\right) = \frac{16}{3}\pi r^3. \quad (1)$$

The total area of the cube is $(2\sqrt{2}r)^3 = 16\sqrt{2}r^3$. Therefore, density of one cube is

$$\frac{\text{volume of spheres}}{\text{volume of cube}} = \frac{\frac{16}{3}\pi r^3}{16\sqrt{2}r^3} \quad (2)$$

$$= \frac{\pi}{3\sqrt{2}} \quad (3)$$

$$= \frac{\pi}{\sqrt{18}}. \quad (4)$$

Since the density of one cube is $\pi/\sqrt{18}$, and the cubes partition Euclidean space, the density of the packing in the limit over all Euclidean space is $\pi/\sqrt{18}$.

Now, we can finally state Kepler's Conjecture.

Conjecture 1.1 (Kepler's Conjecture) *The face centered cubic packing is the densest sphere packing in $\mathbb{R}^3$. The density of this sphere packing is*

$$\frac{\pi}{\sqrt{18}} \approx 0.74048. \quad (5)$$

2 Circle Packing

Circle packing is the 2-dimensional sphere packing problem. We want to know how densely we can pack circles in the plane.

Theorem 2.1 (Thue's Theorem) *The regular hexagonal packing is the densest circle packing in the plane. The density of this circle packing is*

$$\frac{\pi}{\sqrt{12}} \approx 0.90690. \quad (6)$$

The following proof was discovered by Chang and Wang [2]. For the proof, let's think of our circle packing as a point set with each point at least distance 2 apart in Euclidean space. We also need the following definitions and the law of sines.

Definition 2.1 (Convex Sets and Convex Hulls) *A convex set is a set such that for any two points in the set, the line between them is completely contained in the set. A convex hull of a point set P is the smallest convex set containing P .*

Definition 2.2 (Triangulations and Delaunay Triangulations) *A triangulation of a polygon A is a division of A into triangles such that each triangle side is either not adjacent to another triangle or entirely shared by two triangles. A Delaunay triangulation of a point set P , denoted $DT(P)$ is a triangulation of the convex hull of P such that the vertices of the triangles are exactly the points in P , and no point in P is inside the circumscribed circle of any triangle in $DT(P)$ (Figure 6).*

Theorem 2.2 (Law of Sines) *Suppose we have some triangle where a , b , and c are the lengths of the sides and α , β , and γ are the opposite angles. Furthermore, let R be the radius of the triangle's circumscribed circle. Then*

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma} = 2R. \quad (7)$$

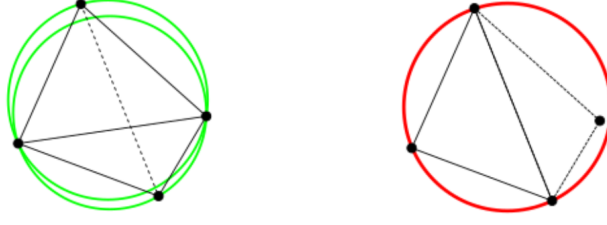


Figure 6: The figure on the left shows a Delaunay triangulation on four points. The figure on the right, however, is not a Delaunay triangulation on the four points because the interior of the circumscribed circle contains a point.

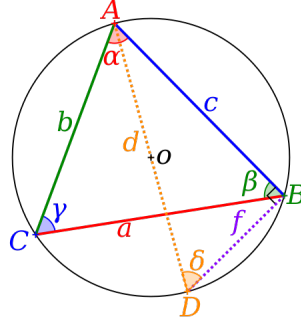


Figure 7: Image for proof of the Law of Sines

Proof of Theorem 2.2: Suppose we have some triangle where a , b , and c are the lengths of the sides and α , β , and γ are the opposite angles (Figure 7). Furthermore, let R be the radius of the triangle's circumscribed circle. We can draw a line segment from A through the origin.

Since the central angle $\angle AOD = \pi$, the angle $\angle ABD = \pi/2$. Therefore, $\triangle ABD$ is a right triangle, and $\sin \delta = c/d = c/2R$. Since γ and δ have the same central angles, $\gamma = \delta$. Therefore,

$$\sin \gamma = \frac{c}{2R} \quad (8)$$

$$\implies 2R = \frac{c}{\sin \gamma}. \quad (9)$$

We can repeat this process for the other two angles. ■

Lemma 2.3 *For every saturated circle configuration, there exists a Delaunay Triangulation [7].*

Proof of Lemma 2.3:

Let a saturated circle configuration $\mathcal{C} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\}$ in the plane be given. We can “lift” each point in the plane onto a paraboloid in $\mathbb{R}^3$ by mapping $\mathbf{p}_i = (x_i, y_i)$ to

$\mathbf{p}_i' = (x_i, y_i, x_i^2 + y_i^2)$. First, notice that circles in the plane are lifted to ellipses in $\mathbb{R}^3$. Suppose such a circle C is defined by the equation,

$$x^2 + y^2 + ax + by + c = 0. \quad (10)$$

Then, since $x^2 + y^2$ gives us the z -coordinate on the paraboloid, we get the following equation for a plane,

$$z = -ax - by - c. \quad (11)$$

Therefore, the circular cylinder that extends through C intersects the paraboloid on a plane, giving us an ellipse. We can compute the convex hull of the circle configuration “lifted” onto the paraboloid. Consider three points that form a face of the convex hull: $\mathbf{p}_i'$, $\mathbf{p}_j'$, and $\mathbf{p}_k'$. I claim that no other point $\mathbf{p}$ in the circle configuration is inside the circumscribed circle of $\mathbf{p}_i$, $\mathbf{p}_j$, and $\mathbf{p}_k$. Suppose that $\mathbf{p}$ was inside the circumscribed circle of $\mathbf{p}_i$, $\mathbf{p}_j$, and $\mathbf{p}_k$. Since these four points can't fall on the same circle in $\mathbb{R}^2$, one of them is farthest from $(0, 0)$. Without loss of generality, suppose $\mathbf{p}_k$ is farthest from $(0, 0)$. Then the plane given by the vertices $\mathbf{p}_i'$, $\mathbf{p}_j'$, and $\mathbf{p}_k'$ would be below the face with vertices $\mathbf{p}_i'$, $\mathbf{p}_j'$, and $\mathbf{p}_k'$, contradicting the fact that the face with vertices $\mathbf{p}_i'$, $\mathbf{p}_j'$, and $\mathbf{p}_k'$ is a face in the convex hull. Therefore, no point $\mathbf{p}$ can be inside the circumscribed circle of three points $\mathbf{p}_i'$, $\mathbf{p}_j'$, and $\mathbf{p}_k'$ that form a face of the convex hull. In conclusion, we can project the edges of the convex hull back into $\mathbb{R}^2$ to give us a Delaunay triangulation of $\mathcal{C}$ (Figure 8).

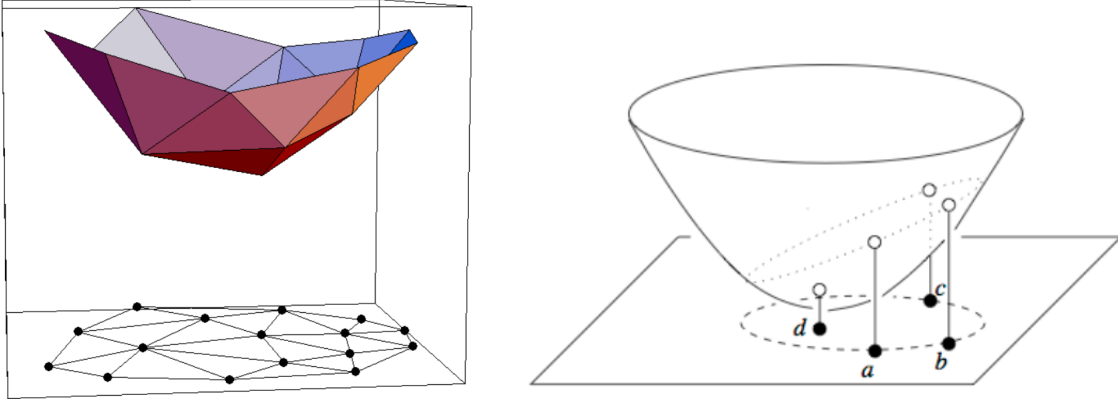


Figure 8: The left image shows how we can find a Delaunay triangulation by projecting points in a plane onto a paraboloid and taking the convex hull of the projected points. The right image depicts that a point is inside a circumscribed circle of three points if and only if it is below the points on the paraboloid [7].

Lemma 2.4 *Let $\mathcal{C}$ be a saturated circle configuration. Let θ be the largest internal angle of a triangle $\triangle ABC$ in a Delaunay triangulation of $\mathcal{C}$. Then*

$$\frac{\pi}{3} \leq \theta < \frac{2\pi}{3}. \quad (12)$$

Proof of Lemma 2.4:

Suppose that $\mathcal{C}$ is a saturated circle packing, and let θ the largest internal angle of a triangle ΔABC in a Delaunay triangulation of $\mathcal{C}$. The first inequality is trivial. The largest internal angle of a triangle is always greater than or equal to $\pi/3$. Otherwise, every internal angle would be less than $\pi/3$, and the sum of the internal angles could not sum to π . Now, we can prove the second inequality by contradiction. Suppose that $\theta \geq 2\pi/3$. Without loss of generality, suppose that A is the smallest internal angle. Since $\theta \geq 2\pi/3$, the other two angles sum to $\pi - 2\pi/3 = \pi/3$. At maximum, the smallest internal angle, A is $\pi/6$; since sine is increasing on $[0, \pi/6]$, $\sin A \leq \sin(\pi/6) = 1/2$. We also have that $\overline{BC} \geq 2$ by our definition of a circle packing. By the law of sines,

$$2R = \frac{\overline{BC}}{\sin A} \geq \frac{2}{\sin A} \geq 4 \quad (13)$$

$$\implies R \geq 2 \quad (14)$$

where R is the radius of the circumscribed circle. Then the center of the circumscribed circle around ΔABC can be added to the circle configuration. The center is at least distance 2 from the points A , B , and C by (14), and there are no other points within distance 2 because the Delaunay triangulation guarantees that there are no other points within the circle (Figure 9).

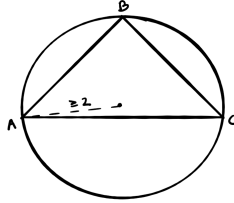


Figure 9: If the radius is greater than or equal to 2, we can add the center point to the circle configuration.

In conclusion, the circle configuration $\mathcal{C}$ is not saturated, which contradicts our hypothesis. $\square$

Lemma 2.5 *Let $\mathcal{C}$ be a saturated circle packing. The density of a triangle ΔABC in a Delaunay triangulation of $\mathcal{C}$ is less than or equal to $\pi/\sqrt{12}$. Equality holds only for a regular triangular with side-length 2 (Figure 10).*

Proof of Lemma 2.5:

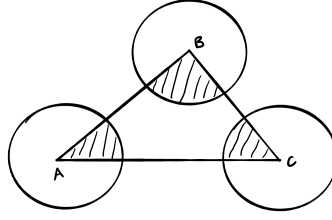


Figure 10: The density of the triangle is the ratio of the shaded area to the area of the whole triangle.

First, notice that the density of $\triangle ABC$ is given by

$$\text{density of } \triangle ABC = \frac{\text{the area of the circles in the triangle}}{\text{the area of the triangle}} \quad (15)$$

$$= \frac{\frac{1}{2}A + \frac{1}{2}B + \frac{1}{2}C}{\text{the area of the triangle}} \quad (16)$$

$$= \frac{\frac{\pi}{2}}{\text{the area of the triangle}}. \quad (17)$$

We get from (15) to (16) because the area of a sector of a circle with angle θ is given by $\frac{\theta}{2\pi} \cdot \pi \cdot r^2 = \frac{\theta}{2} \cdot r^2$ where r is the radius of the circle. For a unit circle, $r = 1$, so the area in the sector is given by $\frac{1}{2}\theta$. We get from (16) to (17) because the sum of the internal angles A , B , and C of a triangle is always π .

Now, without loss of generality, suppose B is the largest internal angle of $\triangle ABC$ (Figure 11). Then by Lemma 2.4,

$$\text{the area of } \triangle ABC = \frac{1}{2} \overline{AB} \cdot \overline{BC} \cdot \sin B \geq \frac{1}{2} \cdot 2 \cdot 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}. \quad (18)$$

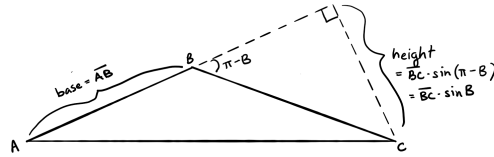


Figure 11: How we compute the area of a triangle.

The inequality (18) holds if and only if $\overline{AB} = \overline{BC} = 2$ and $\sin B = \sqrt{3}/2$ (i.e. $B = \pi/3$), which are necessary and sufficient properties for our triangle to be equilateral. That is, equality holds if and only if $\triangle ABC$ is equilateral with side-length 2. Now, we can substitute the inequality (18) into the denominator of (17) and get that

$$\text{density of } \triangle ABC \leq \frac{\pi}{\sqrt{12}}. \quad (19)$$

Again, equality holds if and only if ΔABC is equilateral with side-length 2. $\square$

Finally, we can complete the proof of Theorem 2.1 (Thue's Theorem). The density of a Delaunay triangulation on any finite point set is given by the weighted sum of the densities of each triangle Δ_i .

$$\text{Density} = \frac{\sum_{\Delta_i} (\text{area of } \Delta_i)(\text{density of } \Delta_i)}{\sum_{\Delta_i} (\text{area of } \Delta_i)} \quad (20)$$

Since the density of each Delaunay triangle is less than or equal to $\pi/\sqrt{12}$, the weighted sum of the densities is also less than or equal to $\pi/\sqrt{12}$. Since the inequality holds for any finite circle packing, it is clear that the limit of the density as the point set grows is also less than or equal to $\pi/\sqrt{12}$. Because equality holds only for regular triangles of side-length 2, a hexagonal lattice is the densest circle configuration (Figure 12). $\blacksquare$

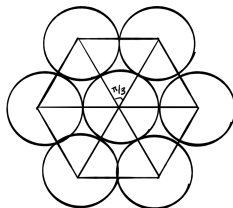


Figure 12: A hexagonal lattice is the densest packing, and it can be constructed from equilateral triangles.

3 Sphere Packing

In 1998, Thomas Hales proved Kepler's conjecture that the FCC packing is the densest packing with density $\pi/\sqrt{18}$. The original proof was about 300 pages [3]. It introduces hundreds of new definitions and lemmas to reduce the problem into one that can be verified with a computer. Specifically, Hales proves Kepler's conjecture by contradiction, reducing all possible counterexamples into finitely many cases and using a computer to show that they cannot exist. Hales uses the HOL Light and Isabelle proof assistants. The software uses an *LCF Approach* to ensure that its deductive reasoning holds. Experts consider a formal proof by HOL Light to be orders of magnitude more reliable than proofs that undergo the standard process of peer review. Since I can't recount the entire proof here, I will only provide an outline.

Proof outline:

First, we must reduce the problem to finitely many spheres. Like the 2-dimensional case, we can consider the density in a part (of a partition) of Euclidean space; if the density is less than $\pi/\sqrt{18}$, then the limit of the density—as we increase the number

of the parts we are considering—is also less than $\pi/\sqrt{18}$. Hales chooses to partition Euclidean space into Marchal cells [8].

Suppose we have a sphere packing and let V be the point set of the centers of the spheres.

Definition 3.1 (Vertices) *A point $\mathbf{v} \in V$ that lies on the boundary of a Marchal cell is called a vertex.*

Definition 3.2 (Edges and Critical Edges) *In a Marchal cell, each line segment between two vertices that lies entirely on the boundary of the cell is called an edge. A critical edge is an edge that is within a certain range of lengths.*

Definition 3.3 (Cell-clusters) *The set of all cells that share a critical edge is called a cell-cluster.*

Definition 3.4 (Dihedral Angles) *Dihedral angles are the angles between two intersecting planes.*

Definition 3.5 (Solid Angles) *A solid angle is the 3-dimensional generalization of a plane angle. The solid angle Ω of a surface S at a point $\mathbf{p}$ is the area of the projection of S onto the unit sphere centered at $\mathbf{p}$. This can be written as*

$$\Omega = \iint_S \hat{\mathbf{n}} \cdot d\mathbf{a} \quad (21)$$

where $\hat{\mathbf{n}}$ is the unit vector that points radially outward from $\mathbf{p}$, and $d\mathbf{a}$ is the differential area of the unit sphere [9].

Definition 3.6 (Hypergraphs) *A hypergraph is a generalization of a graph in which an edge can join any number of vertices. Formally, an undirected hypergraph H is an ordered pair $H = (X, E)$ where X is a set of vertices, and E is a set of nonempty subsets of X , called hyperedges, containing nodes that are adjacent [9].*

A hypergraph can be represented uniquely with a *Levi graph*, which is a bipartite graph with black vertices corresponding to vertices in the hypergraph, and white vertices corresponding to hyperedges (Figure 13) [10].

Definition 3.7 (Hypermaps) *A hypermap is an embedding of a hypergraph on a surface without edge-crossings and with simply-connected faces [10].*

Since the faces are simply-connected, they are homeomorphic to a disk. That is, they can be continuously transformed into a disk without poking holes or creating tears.

Hales relates Kepler’s conjecture to many mathematical inequalities and statements. He encodes nearly 1,000 non-linear inequalities and makes use of the cell-cluster inequality and local annulus inequality, defined below.

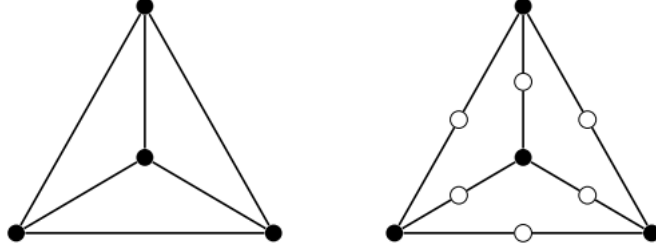


Figure 13: On the left, we have a graph of a tetrahedron. On the right, we have its corresponding Levi graph [10].

Definition 3.8 (Cell Cluster Inequality) *We can assign a real number $\Gamma(\varepsilon, X)$ to each cell X together with one of its critical edges ε . $\Gamma(\varepsilon, X)$ is defined geometrically based on the volume of the cell X , the solid angles at X 's vertices, the dihedral angles along X 's edges, and the lengths of X 's edges. For every critical edge ε with corresponding cell-cluster C ,*

$$\sum_{X \in C} \Gamma(\varepsilon, X) \geq 0. \quad (22)$$

Definition 3.9 (Local Annulus Inequality) *The ball annulus is the set given by $A = \{x \in \mathbb{R}^3 : 2 \leq \|x\| \leq 2.52\}$. Define $f(t) = (2.52 - t)/(2.52 - 2)$, which is a linear function decaying from 1 to 0 moving radially outwards in the annulus. For a countably infinite packing V , since A is closed and bounded, $V \cap A$ must be finite. For a finite packing $V \subset A$, the following inequality must hold.*

$$\sum_{\mathbf{v} \in V} f(\|\mathbf{v}\|) \leq 12 \quad (23)$$

Hales then proves the following logical statement.

$$\forall V (\text{The non-linear inequalities hold.}) \quad (24)$$

$$\text{and } \forall V (\text{The cell-cluster inequality holds.}) \quad (25)$$

$$\text{and } \forall V (V \subset \text{Annulus} \implies \text{The annulus inequality holds.}) \quad (26)$$

$$\implies (\text{Kepler's conjecture holds.}) \quad (27)$$

Hales proves statements (24) and (25) in his paper, but the crux of the proof falls on proving statement (26). Hales proves (26) by contradiction. Suppose that there exists a sphere packing V such that the local annulus inequality is not true, called a *contravening* sphere packing. He proves that each contravening sphere packing gives rise to a hypermap with special properties, called a *tame hypermap*. There are finitely many tame hypermaps that can be enumerated by a computer. Then we can find all contravening sphere packings V associated with each tame hypermap. Lastly, we can

express the conditions on V as a system of non-linear inequalities, relax the nonlinear system to a less restrictive linear system, and show that the system is inconsistent. This shows that V can't exist, which is a contradiction. We can eliminate all possible counterexamples V in this way, showing that (26) is true. ■

4 Higher Dimensions

4.1 Spheres and Volumes in n-Dimensions

Proving the sphere packing problem in 3-dimensions strained human proof abilities, so what about higher dimensions? We can generalize the sphere packing problem to higher dimensions in the following way. A sphere S of radius r centered at $\mathbf{p}$ in $\mathbb{R}^n$ is the set of points distance r from $\mathbf{p}$. This can be written down as $S = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{p} - \mathbf{x}\| = r\}$; one can consider different geometries, but we will focus on Euclidean distance. The volume of an n -dimensional set is defined as the Lebesgue measure of the set. One construction of the Lebesgue measure is as follows [11].

A box B in $\mathbb{R}^n$ is a set of the form

$$B = \prod_{i=1}^n [a_i, b_i] \quad (28)$$

where $\prod$ denotes the Cartesian product, and $-\infty < a_i \leq b_i < \infty$. Then the volume of B is given by

$$\text{vol}(B) = \prod_{i=1}^n (a_i - b_i) \quad (29)$$

where $\prod$ now denotes the standard product. Next, we can define the *outer measure* of $A \subset \mathbb{R}^n$, $\lambda^*(A)$ by

$$\lambda^*(A) = \inf \left\{ \sum_{B \in \mathcal{C}} \text{vol}(B) : \mathcal{C} \text{ is a countable collection of boxes whose union covers } A \right\}. \quad (30)$$

Finally, we define our set A to be Lebesgue measurable if for every subset $S \subset \mathbb{R}^n$,

$$\lambda^*(S) = \lambda^*(S \cap A) + \lambda^*(S \setminus A). \quad (31)$$

If A is Lebesgue measurable, then the Lebesgue measure on A is defined as $\lambda(A) = \lambda^*(A)$.

Intuitively, it is easy to understand the volume of a box in n -dimensions; we just multiply the lengths of the box in every direction. Then we can find the volume of another shape A by filling it with infinitesimally small boxes and summing their volumes.

The explicit formula for the volume of an n -dimensional sphere with radius r is given by

$$V_n(r) = \frac{\pi^{n/2}}{(n-1)!} r^n \quad (32)$$

for any positive integer n [12], but for our purposes, we just need to know that $V_n(r)$ is proportional to r^n .

Proof that $V_n(r) \propto r^n$:

Intuitively, if we have some set K with volume k , and we stretch it by r in every direction, the resulting volume is $r^n k$. The proof is as follows. Let's denote the n -dimensional unit sphere with S^n , and suppose the volume of S^n is k . Now, let S_r^n be the unit sphere stretched by r in each direction (i.e. a sphere of radius r). By definition

$$\text{vol}(S_r^n) = \int_{\mathbf{x} \in S_r^n} d\mathbf{x} \quad (33)$$

where $\mathbf{x} = (x_1, \dots, x_n)$. Now, we can use the change of variables, $\mathbf{x} = T(\mathbf{y}) = r\mathbf{y}$ where $\mathbf{y} = (y_1, \dots, y_n)$. The Jacobian of T is given by a matrix with r 's on the diagonal and zeros everywhere else.

$$D_T = \begin{pmatrix} r & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & r \end{pmatrix}_{n \times n} \quad (34)$$

Since we have a diagonal matrix, the determinant is the product of the diagonal entries.

$$\det |D_T| = r^n \quad (35)$$

Therefore, when we use change of variables, we see that

$$\text{vol}(S_r^n) = \int_{\mathbf{x} \in S_r^n} d\mathbf{x} = \int_{\mathbf{y} \in S^n} r^n d\mathbf{y} = r^n \text{vol}(S^n) = r^n k. \quad \blacksquare \quad (36)$$

Notice that since volume is proportional to r^n , volume grows exponentially as dimensions increase. In high dimensions, the space is unfathomably large, which makes it difficult to search for the best sphere packings by brute force.

4.2 A Simple Lower Bound on Saturated Sphere Packings: 2^{-n}

Consider any saturated sphere packing with spheres of radius r in $\mathbb{R}^n$. Since the packing is saturated, no further spheres can be added without overlap. Then no point in $\mathbb{R}^n$ can be $2r$ away from the centers of all other spheres. This implies that if all the

spheres had radius $2r$, they would cover all space. Since volume is proportional to r^n , if we double the radius, the volume increases by 2^n .

$$V_n(r) = k \cdot r^n \quad (37)$$

$$V_n(2r) = k \cdot (2r)^n = V_n(r) \cdot 2^n \quad (38)$$

where k is some proportionality constant. Therefore,

$$\text{Density} = \frac{\text{volume of spheres}}{\text{volume of space}} \geq \frac{V}{V \cdot 2^n} = 2^{-n} \quad (39)$$

$$\implies \text{Density} \geq 2^{-n} \quad (40)$$

where V is the volume of the spheres. ■

Researchers continually improve bounds on the density of sphere packings. Keith Ball (1992) showed that the density is at least $2n \cdot 2^{-n}$; Stephanie Vance (2011) showed that the density is at least $(6/e)n \cdot 2^{-n}$ when n is divisible by 4; and Akshay Venkatesh (2012) showed that the density is at least $(e^{-\gamma}/2)n \log(\log(n \cdot 2^{-n}))$ for certain sparse sequences of dimensions. For comparison, the best known upper bound is $2^{-0.599n}$, which was proven by Kabatiansky and Levenshtein in 1978 [5].

4.3 Lattice Packings

There is no reason to believe that the densest packings aren't random, but periodic packings (i.e. those with repeating patterns) can come arbitrarily close to the maximum density. Although the best known sphere packings are not always lattices, many good ones are [5]. The following definitions about lattices are from [13].

Definition 4.1 (Lattices) *A lattice in n -dimensions is the span of n vectors with integers as coefficients (Figure 14).*

For example, if we have an n -dimensional basis of vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then the centers of the spheres are the points given by

$$\mathcal{L}(B) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n : a_1, a_2, \dots, a_n \in \mathbb{Z}\} \quad (41)$$

where $\mathcal{L}(B)$ denotes the lattice of the basis B . Equivalently, if we define B as the matrix with columns given by the basis vectors, then

$$\mathcal{L}(B) = \{B\mathbf{x} : \mathbf{x} \in \mathbb{Z}^n\}. \quad (42)$$

If B is a square matrix, the lattice is called a *full rank lattice*; in this paper, we will focus on such lattices. Now, we can define some important properties of lattices.

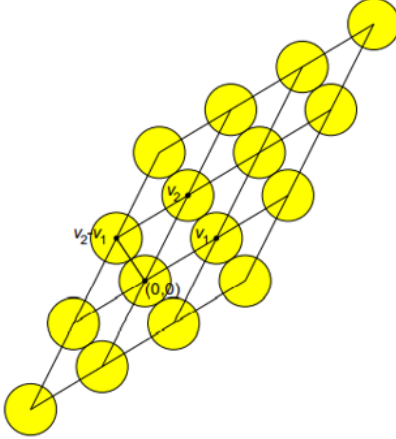


Figure 14: An example of a lattice packing [5].

Definition 4.2 (Fundamental Parrallelepipeds) For any lattice basis B , we define the fundamental parallelepiped $\mathcal{P}(B)$ as follows.

$$\mathcal{P}(B) = \{B\mathbf{x} : \mathbf{x} \in \mathbb{R}^n, \forall i : 0 \leq x_i \leq 1\} \quad (43)$$

That is, $\mathcal{P}(B)$ is the set of all linear combinations of the basis vectors with scalar coefficients in $[0, 1]$.

Definition 4.3 (Lattice Determinants) Let $\Lambda = \mathcal{L}(B)$ be an n -dimensional full-rank lattice with basis B . We define the determinant of Λ as the n -dimensional area of the fundamental parallelepiped $\mathcal{P}(B)$. Therefore,

$$\det(\Lambda) = |\det(B)|. \quad (44)$$

To understand definition 4.3 above, we need to recall a theorem from linear algebra.

Theorem 4.1 (Determinants and Volumes) Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be vectors in $\mathbb{R}^n$, let P be the parallelepiped determined by these vectors, and let A be the matrix with vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ as columns. Then

$$|\det(A)| = \text{vol}(P). \quad (45)$$

Before we begin the proof, recall a characterization of the determinant below.

Definition 4.4 (Determinants) The determinant is a function that maps square matrices to $\mathbb{R}$ with the following properties.

1. Adding some linear combination of the other columns to a column does not change the determinant.
2. Swapping columns multiplies the determinant by -1 .

3. *Multiplying a column by the scalar α multiplies the determinant by α .*
4. *The determinant of the identity matrix is 1.*

Proof of Theorem 4.1:

Let A be a square matrix with vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ as columns, and let P be the parallelepiped determined by these vectors. By definition 4.4, any function that satisfies the four stated properties is the determinant; therefore, $|\det(A)|$ is also uniquely defined by corresponding properties. We can show that the volume of a parallelepiped satisfies the four properties that define $|\det(A)|$; therefore, it is equivalent to the determinant.

1. Without loss of generality, suppose we replace $\mathbf{v}_n$ with a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n-1}$. The volume of the parallelepiped is given by the “base” times the “height.” Here, the base is determined by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n-1}$, and the height is the distance of $\mathbf{v}_n$ orthogonal to all vectors that make up the base. Since every vector in the base is perpendicular to the height, adding a linear combination of them to $\mathbf{v}_n$ won’t change its orthogonal distance from the base. Since our choice of $\mathbf{v}_n$ was arbitrary, replacing a vector with itself plus some linear combination of the other vectors doesn’t affect the volume of the parallelepiped.
2. Swapping two columns just swaps the order of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ without changing P ; therefore, swapping the columns has no effect on $\text{vol}(P)$.
3. Without loss of generality, suppose we scale $\mathbf{v}_n$ by a scalar value of c . This changes the “length” of the vector by c by expanding it in every direction by c . Thus, the perpendicular distance of v_n from the base is also scaled by c . Accordingly the volume changes by $|c|$.
4. Lastly, the columns in the identity matrix are $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$, whose parallelepiped is the n -dimensional unit cube. The unit cube has volume 1.

We have shown that the volume of the parallelepiped P satisfies the four characteristics defining $|\det(A)|$. In conclusion,

$$|\det(A)| = \text{vol}(P). \quad \blacksquare \quad (46)$$

[14]

Definition 4.5 (Unimodular Lattices) *A unimodular lattice is a lattice with determinant 1 or -1 .*

$\mathbb{Z}^n$ is the simplest lattice, but it’s usually not a good packing. For $n \geq 3$, a “checkerboard” packing with offset spheres is typically better. We can define a checkerboard pattern in n -dimensions as follows.

$$D_n = \{(x_1, x_2, \dots, x_n) : x_1 + x_2 + \dots + x_n \text{ is even}\} \quad (47)$$

For such a lattice, the nearest points are of the same distance as $(0, 0, 0, \dots, 0)$ and $(1, 1, 0, \dots, 0)$. Therefore, the minimum distance between points is $\sqrt{2}$. Thus, for a packing D_n , we should consider spheres with radius $\sqrt{2}/2$. D_3 , D_4 , and D_5 are the best-known packings in those dimensions and are provably the best lattice packings. As n grows larger, the holes in D_n grow larger. There are two classes of holes

1. Holes at distance 1 from the lattice like $(1, 0, \dots, 0)$
2. Holes at distance $\sqrt{\frac{n}{4}}$ from the lattice like $(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2})$.
 $\left(\text{distance} = \sqrt{\frac{1}{2} + \dots + \frac{1}{2}} = \sqrt{\frac{n}{4}}\right)$

When $n = 8$, the larger hole is at distance $\sqrt{8/4} = \sqrt{2}$. Since the spheres have radius $\sqrt{2}/2$, we can slip spheres into these holes. In fact, placing spheres in each of these holes gives us another copy of D_8 , doubling the packing density [5]. We can denote the new packing with Λ_8 and formally define it as follows.

Definition 4.6 (Λ_8)

$$\Lambda_8 = \left\{ \mathbf{x} \in \left(\mathbb{Z}^8 \cup \left(\mathbb{Z} + \frac{1}{2} \right)^8 \right) : \sum_{i=1}^8 x_i \equiv 0 \pmod{2} \right\} \quad (48)$$

If we want to consider unit spheres instead of spheres with radius $\sqrt{2}/2$, we can consider the lattice E_8 defined by $E_8 = \frac{1}{\sqrt{2}}\Lambda_8$ [15].

Similarly, mathematicians have discovered an incredibly dense packing in $\mathbb{R}^{24}$, called the Leech-lattice, denoted Λ_{24} .

Definition 4.7 (The Leech-lattice) *The Leech-lattice, denoted Λ_{24} , is the unique lattice in $\mathbb{R}^{24}$ with the following properties.*

1. *It is unimodular.*
2. *It is even (the square of the length of each vector in Λ_{24} is an even integer).*
3. *The length of every nonzero vector in Λ_{24} is at least 2.*

The last condition ensures that unit spheres centered at the lattice points don't overlap. The Leech-lattice is an extremely symmetrical and dense packing. Each sphere is tangent to 196,560 neighboring spheres, and there is no room to move any of the balls [16].

In $\mathbb{R}^8$ and $\mathbb{R}^{24}$, E_8 and the Leech-lattice, respectively, are extremely symmetrical and dense packings. In 2016, Maryna S. Viazovska used modular forms to prove that these were optimal packings in their respective dimensions [15],[17]. The optimal packings for all other dimensions are still unknown.

Viazovska takes advantage of *linear programming bounds* to find an upper bound for the density of sphere packings. These bounds were developed by Cohn and Elkies [18],[19],[20], and they make use of certain auxiliary functions to obtain bounds. The theory, however, does not provide information about how to choose an auxiliary function to make the bounds sharp (Figure 15). Viazovska constructs an auxiliary function that proves the density of E_8 and the Leech lattice are maximal. To define linear programming bounds, we need the following definitions.

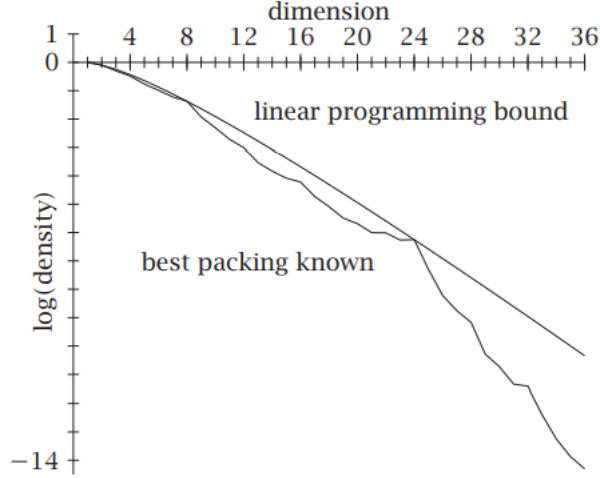


Figure 15: The logarithm of sphere packing density as a function of dimension. The upper curve is the numerically optimized linear programming bound, and the lower curve is the best-known sphere packing. The densest packings fall somewhere between the two curves. Notice, prior to Viazovska’s work, the best-known linear programming bounds and the densities of E_8 and the Leech lattice were amazingly close, but not identical [17].

Definition 4.8 (Fourier Transform) Define the Fourier transform $\hat{f}$ of an integrable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$\hat{f}(\mathbf{y}) = \int_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \langle \mathbf{x}, \mathbf{y} \rangle} d\mathbf{x} \quad (49)$$

where $\langle x, y \rangle$ denotes the standard inner product (the dot product) in $\mathbb{R}^n$.

Definition 4.9 (Big-“O” Notation) $f(x) = O(g(x))$ as $x \rightarrow \infty$ if there exists $k > 0$ such that $\frac{|f(x)|}{|g(x)|} < k$ for all $x > a$ for some $a \in \mathbb{N}$.

Definition 4.10 (Schwartz Functions) A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Schwartz function if f is infinitely differentiable and

$$f(\mathbf{x}) = O((1 + |\mathbf{x}|)^{-k}) \quad (50)$$

for all $k \in \mathbb{N}$, and the same holds for all partial derivatives of f of every order.

Theorem 4.2 (Cohn and Elkies) *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a Schwartz function and r be a positive real number such that $f(\mathbf{0}) = \hat{f}(\mathbf{0}) > 0$, $\hat{f}(\mathbf{y}) \geq 0$ for all $\mathbf{y} \in \mathbb{R}^n$, and $f(\mathbf{0}) \leq 0$ for $|\mathbf{x}| \geq r$. Then the sphere packing density in $\mathbb{R}^n$ is at most $V_n(r/2)$ —the volume of a ball with radius $r/2$ in n -dimensions.*

The challenge comes in choosing f to minimize r . Viazovska discovered an integral transformation, which allowed her to construct an optimal f .

5 Conclusion

The sphere packing problem is notoriously a simply-stated problem with intuitive answers but excruciatingly difficult proofs. Humans have considered the problem for centuries but have only found the answer in 1, 2, 3, 8, and 24 dimensions. In this paper, we began with an overview of the history and applications of sphere packing. In signal processing, we saw that high-dimensional sphere packings are used to restrict the vocabulary of sent signals and reduce the ambiguity in received signals. Next, we rigorously defined the sphere packing problem in two and three dimensions. In two dimensions, we saw a proof of Thue’s theorem, showing that the maximum density of a 2-dimensional circle packing is $\pi/\sqrt{12}$ and that we can achieve this density with a hexagonal circle packing. Then, we reviewed Hales’ Theorem, which proved that the maximum density in three dimensions is $\pi/\sqrt{18}$. Such a density can be achieved with the face centered cubic (FCC) packing. We reviewed how Hales’ computational proof works.

Afterwards, we generalized the sphere packing problem to higher dimensions. Spheres in n dimensions are sets of points that are equidistant from some center point, and the volume of an n -dimensional sphere is given by the Lebesgue measure. In n -dimensions, we provide some lower bounds on the packing density. Then we discuss lattice constructions of sphere packings. Such constructions give us many of the best-known packings. In fact, researchers have been able to create extremely symmetrical lattice packings in 8 and 24 dimensions, known as the E_8 lattice and the Leech lattice, respectively. In 2016, these lattices were proven to have maximum density in their respective dimensions. We discussed intuitively that the proof works by finding an auxiliary function to make linear programming bounds sharp.

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